Fourth Biweekly Report (May 18 – May 31, 2026)

Reporting Period: May 18 – May 31, 2026 (Weeks 10–11) | Submission: May 31, 2026

1. Team Summary — WBS Progress Overview

This report covers Weeks 10–11 (May 18–31, 2026), during which HeartCompass transitioned from core development into integration, validation, and delivery readiness. The team completed all remaining WP3 tasks (WP3.3–WP3.4 at 100%), finalized WP4 Channel Integration (WP4.1–WP4.3 at 100%, WP4.5-4.6 at 95%), and brought WP5 CLI/Deployment to full operational capability (WP5.1–WP5.2 at 100%, WP5.3 at 95%). WP3.5 coherence testing reached 100% pass rate; WP3.6 prompt optimization achieved 80% with measurable quality gains. Total: 220 hours across 12 tracked WBS tasks, weighted completion 94.8%. All milestones on track.

1.1 WBS Task Progress Matrix (Weeks 10–11)

WBS Task	Description	Owner	Hours	Compl.	Status
WP2.1-2.5	Persona Building Pipeline (Stable)	Leishiyu/Haojie	—	**100%**	**Complete**
WP3.1-3.2	ConversationGraph Core & Memory Trimming	Yihan Liu	38	**100%**	**Complete**
WP3.3	LLM Call with Retry Strategy	Yihan Liu	14	**100%**	**Complete**
WP3.4	Feed Sync — Conversation-to-Persona Loop	Yihan Liu	12	**100%**	**Complete**
WP3.5-3.6	Coherence Testing & Prompt Optimization	Yihan Liu	32	**90%**	**In Progress**
WP4.1	Lark WebSocket Connection & Message Routing	Leishiyu Chen	8	**100%**	**Complete**
WP4.2	Bot Command Menu & Persona Interaction	Tian/Leishiyu	22	**100%**	**Complete**
WP4.3	Message Batching & Buffer Management	Tian/Leishiyu	14	**100%**	**Complete**
WP4.5-4.6	WebSocket Testing & Card Template System	Leishiyu Chen	22	**95%**	**In Progress**
WP5.1	CLI Framework: Doctor, Setup, Logs Commands	Tian Bu	16	**100%**	**Complete**
WP5.2	CLI Auth Login & Session Persistence	Tian/Haojie	16	**100%**	**Complete**
WP5.3-5.4	Docker, Alembic, Deployment & Integration Tests	Haojie Hu	26	**84%**	**In Progress**

1.2 Overall Assessment & Outlook

Overall: WP2 Persona Building maintained stable operation across all 4 dimensions with zero regressions. WP3 Conversation System achieved full feature completeness: LLM calls with exponential-backoff retry (WP3.3, 100%), 3-tier confidence-scored feed sync closing the conversation-to-persona loop (WP3.4, 100%), comprehensive 20-turn coherence test suite at 100% pass rate across all 5 FigureRole types (WP3.5, 100%). Prompt optimization hit 80% with 35% repetition reduction and 28% naturalness gain. WP4 Channel Integration is production-ready: WebSocket with heartbeat, graceful shutdown, and live-sandbox-validated reconnection; complete 4-command bot menu with rate limiting; message batching with 50-msg cap and cross-device dedup; card templates with null-safe rendering. WP5 CLI at 100% (doctor/setup/logs, auth login with JWT persistence + transparent refresh). Docker Compose with healthcheck-gated orchestration and Alembic migration 003 validated staging-ready (95%). WP5.4 integration test plan approved for Week 12. Risk: LOW. Fully on schedule.

1.3 Effort Distribution by WBS Area

WP2 (Persona + Recall Enh.): 16h (7.3%) | WP3 (Conversation): 96h (43.6%) | WP4 (Channel): 70h (31.8%) | WP5 (CLI + Deploy + Gov): 38h (17.3%)

WP3 Conversation System	96h	96h
WP4 Channel Integration	70h	70h
WP5 CLI + Deploy + Gov	38h	38h
WP2 Persona + Recall	16h	16h

1.4 Weekly Milestone Achievement

Week	Key Deliverables (WBS)	Description	Status
Week 10 (May 18–24)	WP3.3–WP3.5 + WP4.2–WP4.5 + WP5.1–WP5.3	LLM retry complete (100%), feed sync 100%, persona commands interactive with null-safe cards, buffer overflow protected, WebSocket reconnection tested, hybrid search integrated, Docker Compose with healthcheck gating, CLI auth login 95%, 20-turn coherence scenarios.	**Achieved**
Week 11 (May 25–31)	WP3.5–WP3.6 + WP4.5–WP4.6 + WP5.2–WP5.4	Coherence suite 100% pass rate after topic-segmented summary fix, prompt optimization 80% (35% repetition reduction, 28% naturalness gain), WebSocket live sandbox validated, card templates at 90%, CLI auth 100%, Alembic migration validated, test plan approved.	**On Track**

1.5 Key Risks & Issues Log

- **Buffer overflow in message batching (resolved, May 22):** Per-user buffer grew unbounded at 50+ msg/15s. Fixed with 50-message hard cap + auto force-flush.
- **LLM API rate limiting under concurrency (resolved):** 3+ concurrent ConversationGraph instances triggered 429 errors. Mitigated by WP3.3 exponential-backoff retry (max 3 attempts, 1s/2s/4s delays). Request queue planned.
- **Card template empty-field display regression (resolved, May 23):** WP4.2 fix caused 'None' literal rendering for empty fields. Corrected via null-safe accessors with per-dimension DEFAULT_PLACEHOLDERS. ~0.5 day rework.

Leishiyu Chen

RBS Area: Data Layer & Channel Integration (RBS R1.2.1, R2.2.1) | Total Hours: 48h

I. WBS Task Breakdown

WBS Task	Description	Hours	Compl.	Status
WP4.1	WebSocket Connection & Message Routing	8	**100%**	**Complete**
WP4.2	Bot Command Menu — Persona Interaction	12	**100%**	**Complete**
WP4.3	Message Batching & Buffer Management	6	**100%**	**Complete**
WP4.5	WebSocket Reconnection & Error Recovery Tests	14	**100%**	**Complete**
WP4.6	Card Template System & Display Refinement	8	**90%**	**In Progress**

II. Period Summary

My primary focus in Weeks 10–11 was completing the WP4 Channel Integration pipeline end-to-end. WP4.1 WebSocket connectivity reached 100% with open_id-based user-to-FR mapping, 30s heartbeat keepalive, and graceful shutdown. WP4.2 delivered the complete bot command suite at 100%: /menu, /list_available_persons, /show_persona with null-safe accessors, and /build_persona with async pipeline triggering. WP4.3 message batching was hardened with 50-message cap, force-flush, and cross-device content-hash dedup (100%). WP4.5 WebSocket reconnection testing reached 100% after live Lark sandbox validation. WP4.6 card template refinement at 90%: null-safe rendering, text truncation, error-state templates, and 4-stage build-progress indicators. Total: 48h.

III. Key Progress by WBS Task

- **[WP4.2 · 12h · 100%] Bot Command Menu — Complete Persona Interaction Suite** Completed all 4 Lark Bot commands at 100%. /menu displays available commands with usage hints. /list_available_persons queries user FR records with ID, name, type, and last-updated timestamp. /show_persona renders structured persona cards across 4 core dimensions with null-safe accessors resolving the 'None' display regression via per-dimension Chinese DEFAULT_PLACEHOLDERS. /build_persona triggers the async FRBuildingGraph pipeline with progress-tracking card. All commands include input validation, Chinese error messages, and rate limiting (5 req/30s per user).
- **[WP4.3 · 6h · 100%] Message Batching — Overflow Protection & Hardening** Hardened batching with production safeguards: 50-message hard cap per user buffer (exceeded → force-flush + WARNING log), suppressed duplicate timer firings during in-progress flush, cross-device deduplication via SHA256(open_id + text + timestamp/15s bucket) with TTL-limited hash set. Correctly handles rapid-fire bursts, single-message turns, timer/send races, and multi-device overlap across desktop and mobile.
- **[WP4.5-4.6 · 22h · 95%] WebSocket Testing & Card Template Refinement** Built comprehensive WebSocket test suite: connect/disconnect cycle validation, exponential-backoff reconnection (1s/2s/4s, max 3 attempts), message queuing with replay on reconnect, invalid auth rejection, and 30s idle timeout proactive reconnect. Week 11 milestone: completed live Lark sandbox integration validation (WP4.5 at 100%). Card templates refined with null-safe accessors using contextual Chinese placeholders, 500-char truncation with 'Show More' expansion, 4-stage build-progress indicator, and 3 error-state templates. Dark-mode palette pending.

IV. Challenges & Solutions

Issue: Empty-field card regression & cross-device message duplication (both resolved)

→ WP4.2 fix caused 'None' rendering for empty MEMORY fields → added null-coalescing with per-dimension DEFAULT_PLACEHOLDERS. Users switching desktop/mobile within 15s batching window produced near-duplicate messages → content-hash dedup with 15s-bucketed timestamps. Verified across 50+ cross-device test sessions.

V. Key Deliverables

✓ Production-ready WebSocket layer with heartbeat and graceful shutdown (100%)
✓ Complete bot command menu: /menu, /list, /show_persona, /build_persona (100%)
✓ Hardened batching: 50-msg cap, force-flush, timer race protection, cross-device dedup (100%)
✓ Card template system: null-safe accessors, 4-stage progress, error-state templates (90%)

VI. Next Steps

→ WP4.6: Complete dark-mode palette; add /delete_persona and /update_persona commands
→ Support WP5.4 channel-layer regression testing with automated WebSocket message replay
→ Prepare WP4 handoff: WebSocket lifecycle, bot command API, card template schema

Yihan Liu

RBS Area: Conversation & Dialogue System (RBS R2.1) | Total Hours: 52h

I. WBS Task Breakdown

WBS Task	Description	Hours	Compl.	Status
WP3.3	LLM Call with Retry Strategy	6	100%	Complete
WP3.4	Feed Sync — Conversation-to-Persona Loop	8	100%	Complete
WP3.5	Multi-Turn Coherence Testing & Optimization	20	100%	Complete
WP3.6	Prompt Template Refinement & Tone Tuning	12	80%	In Progress
WP3.2f	Memory Trim Boundary Condition Fix	6	100%	Complete

II. Period Summary

My work in Weeks 10–11 completed the WP3 Conversation System to full feature readiness and advanced into systematic testing and quality optimization. WP3.3 LLM call reached 100% with production-grade exponential-backoff retry and structured output validation. WP3.4 feed sync achieved 100% with 3-tier confidence-scored update logic — closing the critical conversation-to-persona feedback loop. WP3.5 delivered a comprehensive 20-turn coherence test suite covering all 5 FigureRole types, reaching 100% pass rate (20/20 scenarios) after implementing topic-segmented conversation summaries. WP3.6 prompt tone optimization reached 80% with 35% repetition reduction and 28% naturalness gain. Total: 52h.

III. Key Progress by WBS Task

- **[WP3.4 · 8h · 100%] Feed Sync — Conversation-to-Persona Closed Loop** Completed feed sync node closing the critical feedback loop. 3-tier confidence threshold: (1) ≥0.8 → direct FR core field update with full audit trail; (2) 0.5–0.79 → upsert as FineGrainedFeed with PENDING_REVIEW status; (3) <0.5 → observation log for manual curation. Integrated with WP2.3 conflict detection: contradictory high-confidence insights automatically create CONFLICT_PENDING entries preserving both values. Sync latency <2s per turn. Validated across 12 synthetic scenarios covering all 5 FigureRole types.
- **[WP3.5 · 20h · 100%] Multi-Turn Coherence Testing — 100% Pass Rate Achieved** Built comprehensive test suite: parametrized 5/10/20-turn scenarios across all 5 FigureRole types (family/friend/mentor/colleague/partner). Validates response relevance, persona consistency, context retention, and memory trim correctness. Edge cases: rapid topic switching, 500+ word monologue, Chinese/English code-switching. Reached 100% pass rate (20/20 scenarios) in Week 11 after implementing topic-segmented conversation summaries that prevent emotional-framing carryover between unrelated topics — resolving the partner-role context bleed bug.
- **[WP3.6 · 12h · 80%] Prompt Template Refinement — Tone & Anti-Repetition** Analyzed 250+ LLM responses identifying 3 core issues: over-formal language in friend/partner roles, repetitive phrases (3–4x per 10 turns), and underuse of persona-specific speech patterns. Phase-1 delivered: role-specific formality parameters, anti-repetition instructions with phrase-history tracker, and 3–5 concrete speech examples per persona. Quantitative results: 35% reduction in repetition frequency (3.2→2.1 avg. repeats/10 turns), 28% improvement in perceived naturalness (blind A/B test, n=40). Phase 2 (emotion-aware tone modulation, cultural context adaptation) scoped for Week 12.

IV. Challenges & Solutions

Issue: Rapid topic switching causing emotional-framing context bleed (resolved, Week 11)

→ Partner-role 20-turn test with 5 rapid topic switches showed emotional carryover from 'work stress' into 'weekend plans'. Root cause: rolling summary condensed all topics into one undifferentiated paragraph. → Implemented topic-segmented summary with embedding-based semantic shift detection — each topic gets labeled paragraph, prompt attends primarily to most recent segment. 100% pass rate achieved.

V. Key Deliverables

✓ LLM call node with exponential-backoff retry (max 3) and structured output validation (100%)
✓ Feed sync node: 3-tier confidence threshold integrated with WP2.3 conflict system (100%)
✓ 20-turn coherence test suite: 20 scenarios, 100% pass rate, all 5 FigureRole types (100%)
✓ Phase-1 prompt optimization: anti-repetition (35% reduction), formality tuning (80%)

VI. Next Steps

→ WP3.6: Phase-2 — emotion-aware tone modulation, cultural context adaptation (Week 12)
→ Add 30/50-turn extended scenarios for long-session stability testing
→ Prepare WP3 handoff: architecture doc, prompt engineering guide, test coverage report

Haojie Hu

RBS Area: Knowledge Base, Vector Recall & Deployment (RBS R1.2.2, R2.1.1, R3.1.2) | Total Hours: 48h

I. WBS Task Breakdown

WBS Task	Description	Hours	Compl.	Status
WP2.4e	Hybrid Search — Semantic + Keyword Recall	10	**100%**	**Complete**
WP3.1o	Recall Dimension Weight Tuning for Conversation	6	**100%**	**Complete**
WP5.2	CLI Auth Login & Session Persistence	10	**100%**	**Complete**
WP5.3	Docker Compose, Alembic & Deployment Config	16	**95%**	**In Progress**
WP5.4	System Integration Testing — Test Plan Design	6	**50%**	**In Progress**

II. Period Summary

My work in Weeks 10–11 spanned recall enhancements, deployment infrastructure, and system testing preparation. WP2.4 hybrid search combined semantic vector recall with BM25 keyword scoring via Reciprocal Rank Fusion — 18–22% recall improvement for named entities and mixed-language queries (10h, 100%). WP3.1 optimization tuned per-dimension recall weights (MEMORY 1.2, PROCEDURAL_INFO 1.1, PERSONALITY 0.9, INTERACTION_STYLE 0.8), reducing LLM input tokens ~12% (6h, 100%). WP5.2 CLI auth login completed to 100% with JWT persistence, transparent refresh, and token revocation. WP5.3 delivered Docker Compose with custom healthcheck-gated pgvector+app orchestration and Alembic migration 003 with verified rollback (16h, 95%). WP5.4 integration test plan with 8 critical-path scenarios approved for Week 12 (6h, 50%). Total: 48h.

III. Key Progress by WBS Task

- **[WP2.4-enh · 10h · 100%] Hybrid Search — Semantic + Keyword Recall Fusion** Implemented hybrid search engine combining semantic vector recall with BM25 keyword scoring via Reciprocal Rank Fusion (k=60). Pipeline: query vectorization → top-K×2 semantic recall via pgvector HNSW index → parallel BM25 keyword search (PostgreSQL to_tsvector/to_tsquery with language stemming) → RRF merge. Evaluation results: 18% recall@5 improvement for mixed Chinese-English queries, 22% for named entities, 15% for rare terms. RRF merge latency <50ms for top-20 candidate pools. Validated on 200-query benchmark covering 5 semantic categories.
- **[WP5.2 · 10h · 100%] CLI Auth Login — JWT Flow & Session Management** Completed CLI auth subsystem to 100%. Full implementation: session file storage (~/.immortality/session.json, 0600 permissions, atomic temp-file write with O_CREAT|O_EXCL), transparent token refresh (401 response → refresh → retry original request), `immortality whoami` command (user identity + token expiry display), logout (secure local deletion + server-side token revocation). Security guarantees: tokens never logged, file permissions enforced at OS level, refresh rotation verified across 3 consecutive cycles. Co-developed with Tian Bu.
- **[WP5.3-5.4 · 22h · 84%] Docker Deployment & Integration Test Planning** Docker Compose infrastructure: pgvector/pgvector:pg16 with custom healthcheck (verifies both pg_isready AND vector type availability in pg_available_extensions → resolves ~8s pgvector compilation loading race) + app container with multi-stage Dockerfile and depends_on service_healthy. Alembic migration 003 adds source_type and trigger_event columns to FROverallUpdateLog; auto-runs on container startup; downgrade path verified. Staging validation: 30 cold-start cycles, zero failures, ~10s average startup time. WP5.4 test plan identifies 8 critical-path scenarios (onboarding→persona→conversation, multi-session evolution, concurrent graph execution, WS disconnect/reconnect, DB pool under 20 concurrent, LLM failure recovery, Docker restart state recovery, cross-platform delivery). Approved for Week 12 execution with detailed specifications.

IV. Challenges & Solutions

Issue: Docker pgvector loading delay causing healthcheck false negatives (resolved)

→ pgvector compilation took ~8s during first run; default pg_isready passed before extension availability, causing CREATE EXTENSION failures. → Custom healthcheck verifying both pg_isready AND vector type in pg_available_extensions. Extended start_period to 15s. 30 cold-start cycles in staging — zero failures.

V. Key Deliverables

✓ Hybrid search engine: semantic + BM25 keyword recall with RRF fusion (18-22% recall gain, 100%)
✓ Dimension weight tuning: MEMORY 1.2, PROCEDURAL 1.1, PERSONALITY 0.9, STYLE 0.8 (100%)
✓ CLI auth login: JWT persistence, transparent refresh, session status, token revocation (100%)
✓ Docker Compose: pgvector/pg16 + app, custom healthcheck, Alembic 003 with rollback (95%)

VI. Next Steps

→ WP5.3: Create docker-compose.prod.yml with resource limits and restart policies
→ WP5.4: Execute critical-path scenarios 1-4 manually; automate 5-8 with test scripts
→ Prepare WP5 handoff: deployment runbook, migration guide, healthcheck configuration

Tian Bu (Team Leader)

RBS Area: CLI, Bot Commands & Project Governance (RBS R2.2.2, R3.1.1) | Total Hours: 44h

I. WBS Task Breakdown

WBS Task	Description	Hours	Compl.	Status
WP5.1	CLI Framework — Doctor, Setup, Logs Commands	10	**100%**	**Complete**
WP5.2	CLI Auth Login — Co-development	6	**100%**	**Complete**
WP4.2	Bot Commands — Co-development & Integration	8	**100%**	**Complete**
WP4.3	Message Batching — Co-development & Review	4	**100%**	**Complete**
WP5.3	Docker & Deployment — Oversight & Validation	4	**95%**	**In Progress**
—	Project Governance & Biweekly Report	12	—	**Ongoing**

II. Period Summary

My work in Weeks 10–11 spanned CLI completion, bot command co-development, deployment oversight, and project governance. WP5.1 CLI framework reached 100%: doctor validates Python version, env vars, dependencies, DB connectivity, and Lark credentials; setup provides interactive .env generation with masked password input; logs supports --level/--module/--tail filtering with --json output. WP5.2 auth login co-implemented to 100% — I designed the CLI UX (secure getpass, session file 0600, automatic Authorization header injection). WP4.2/4.3 bot commands and batching co-developed with Leishiyu Chen. WP5.3 deployment oversight: validated healthcheck gating, approved migration 003, verified staging. Governance (12h): maintained WBS tracking through Weeks 8–9, normalized logging, coordinated 14-merge Week 10 with zero conflicts, tagged v0.4.0-dev (May 24), compiled this report. Total: 44h.

III. Key Progress by WBS Task

- **[WP5.1 · 10h · 100%] CLI Framework — Full Feature Completion (75% → 100%)** Brought immortality CLI from 75% to 100% feature completion. Doctor validates Python 3.11–3.13, all required env vars (DATABASE_URL, LARK_APP_ID/SECRET, ARK_API_KEY/BASE_URL), dependency imports, DB connectivity, and Lark API credentials with clear pass/fail indicators. Setup provides interactive .env generation with masked getpass input, SQLAlchemy table creation, and pgvector extension verification. Logs supports --level/--module/--tail/--follow filters plus --json structured output for pipeline consumption. Consistent UX throughout: colored status indicators, progress spinners, explicit exit codes (0/1/2) for CI/CD scripting integration.
- **[WP4.2/4.3 · 12h · 100%] Bot Commands & Message Batching — Co-development to Production** Co-developed the complete Lark Bot interaction layer with Leishiyu Chen. WP4.2 contributions: command routing dispatcher with fuzzy Chinese/English matching, /list_available_persons implementation, and error response templates for invalid or malformed commands. WP4.3 contributions: 15s configurable wait timer with per-user buffer isolation (keyed by open_id), timer-reset-on-new-message logic, and design review of the 50-message hard cap and cross-device deduplication mechanism. All 4 bot commands plus the batching pipeline verified end-to-end on both Lark desktop client and mobile app. Both subsystems at 100%, production-hardened with comprehensive error handling.
- **[Governance · 12h · Ongoing] Project Governance, Release Management & Biweekly Report** Maintained WBS milestone tracking across Weeks 8–9: Week 8 checkpoint (May 16) recorded WP2.1-2.5 and WP3.1-3.2 at 100%, WP3.3 at 80%, WP3.4 at 60%, WP4.1 at 100%, WP4.2 at 85%, WP4.3 at 70%, WP5.1 at 85%, WP5.2 at 50%, WP5.3 at 40%. Week 9 checkpoint (May 23): WP3.3-3.4 reached 100%, WP4.2-4.3 hardened to 100%, WP5.2 advanced to 60%, WP5.3 to 50%. Normalized logging format across all ConversationGraph nodes for consistency. Coordinated Week 10 merge cadence: 4 parallel feature branches, 14 --no-ff merges, morning-merge discipline (all merges before 11:00 AM CST) achieving zero unresolved conflicts reaching main. Tagged v0.4.0-dev on May 24 marking Weeks 8–9 deliverable completion. Compiled this fourth biweekly report: 12 WBS tasks, 220 total hours, 94.8% weighted completion across all work packages. Risk: LOW, all milestones on schedule.

IV. Challenges & Solutions

Issue: Coordinating 4 parallel branches with 14 merges across 7 days (managed)

→ Week 10 saw peak velocity: all 4 members committing simultaneously → elevated merge conflict risk. → Morning-merge cadence: branches merged before 11:00 AM CST, afternoon for resolution and integration testing. All merge commits carry team leader signature for auditable integration. Zero unresolved conflicts reached main.

V. Key Deliverables

✓ immortality CLI v1.0: doctor (5 checks), setup (interactive .env + DB), logs (filtered + --json)
✓ CLI auth login: secure getpass, JWT persistence, auto-refresh, logout with revocation (100%)
✓ Lark Bot production suite: 4 commands with validation, Chinese errors, rate limiting (100%)
✓ v0.4.0-dev release (May 24); Fourth biweekly report: 12 tasks, 220h, 94.8% completion

VI. Next Steps

→ WP5.3: Finalize docker-compose.prod.yml; production deployment dry-run (Week 12)
→ Coordinate WP5.4 test execution: assign scenarios, track pass/fail
→ Prepare final presentation: architecture, WBS milestones, key metrics, demo outline